HepatoTwin- Weekly Action Plan

Meeting-by-Meeting Plan

Meeting	Date	Task 1 (twin state + evidence)	Task 2 (optimizer + recipes)
1	Fri 21 Aug	NHANES data downloaded and linked; repository setup started	Recipe resources identified; database access/schema investigation started
2	Fri 28 Aug	EDA of CAP/steatosis; candidate feature list; preprocessing pipeline started	Read RecipeDB/FlavorDB/SpiceRx/DietRx papers; initial understanding of recipe schema and ingredient extraction
3	Thu 3 Sept	Train first baseline model using NHANES with train/test split	Build robust RecipeDB parser and extract recipe metadata, ingredients, cuisine, and preparation information
4	Fri 11 Sept	Compare baseline models and select initial patient-state prediction approach	Define a structured internal recipe representation and begin nutrient profiling of parsed recipes
5	Fri 18 Sept	Finalize initial predictive model and evaluate performance	Build recipe nutritional feature profiles: calories, macronutrients, fat-related variables, etc.
No meeting as it was mid-sem week	Fri 25 Sept	Light week: review model results and identify limitations	Light week: test parser and recipe representation on a larger recipe batch
6	Thu 1 Oct	Begin calibration and SHAP/explainability; draft evidence table from literature	Define recipe evaluation criteria: health alignment, cuisine compatibility, and practical feasibility
7	Fri 9 Oct	Complete initial evidence table and connect patient characteristics to nutritional objectives	Implement individual scoring functions for recipe-level criteria

8	Fri 16 Oct	Build Layer 3 logic: patient state → relevant nutritional priorities → priority levels	Implement patient-specific health alignment scoring for recipes
9	Fri 23 Oct	Integration checkpoint met early (end-to-end pipeline and demo run). Build Layer 5 scenario engine: Recipe A vs B → evidence-graded projected direction (§8). Save the current Layer 4 evaluation as the “before” run, then download DR1IFF and FPED and add the patient’s habitual diet from NHANES 24-hour recall so priorities are personal (§17.1). Add new evidence rows: coffee, fasting, null findings (§17.13)	Recommendation diversity re-ranking and 7-day meal plan with weekly constraints (§17.6). Add rating-based acceptability (§17.12). Start recruiting 5–10 pilot evaluators
10	Fri 30 Oct	Explainability and safety: fibrosis risk (FIB-4) validated against measured liver stiffness, plus referral pathway (§17.2); MASLD/MetALD labelling and safety guardrails (§17.3); per-patient SHAP and conformal CAP intervals (§17.4); causal knowledge graph of 10–15 links (§5.3); download NHANES III ultrasound and Linked Mortality Files and map variables (§17.9)	Explainability: knowledge-graph-traced recipe explanations (§7.3); grounded plain-language summaries with automatic faithfulness check (§17.7); improve cuisine score (ingredient co-occurrence) and feasibility score (process/utensils) (§7.1)
11	Fri 6 Nov	FastAPI backend wrapping Layers 2–5 with an SQLite database	React dashboard: patient state, priorities with evidence, side-by-side recipe comparison, scenarios and

		(§16); re-assessment timeline storing dated measurements (§17.5); simple 3D liver S0–S3 with timeline slider (§9)	summaries (§10). New recipe criteria: ultra-processing (FoodProX, §17.10) and glycaemic load (§17.11)
12	Fri 13 Nov	Evaluation: NHANES III external and mortality validation (§17.9); fibrosis triage accuracy; conformal coverage; fairness audit by sex, ethnicity and age (§17.8, §12.4)	Evaluation: re-run Layer 4 evaluation with new criteria (personalisation, diversity, meal-plan constraints, summary faithfulness); run pilot evaluation with 5–10 people (§12.2, §12.4)
13	Fri 20 Nov	Freeze scope; consolidate results; update Appendix A and write the twin-state, evidence and scenario chapters. If behind, apply the proposal's cut list (3D fidelity first, then panel size, then evidence-table breadth)	Freeze scope; consolidate results; update Appendix B and write the ranking-engine and evaluation chapters; prepare internal evaluation
14	Fri 27 Nov	Final debugging, documentation and presentation preparation; reproducibility check (one command rebuilds data, models and results)	Final debugging, documentation and presentation preparation; live demo rehearsal of the dashboard